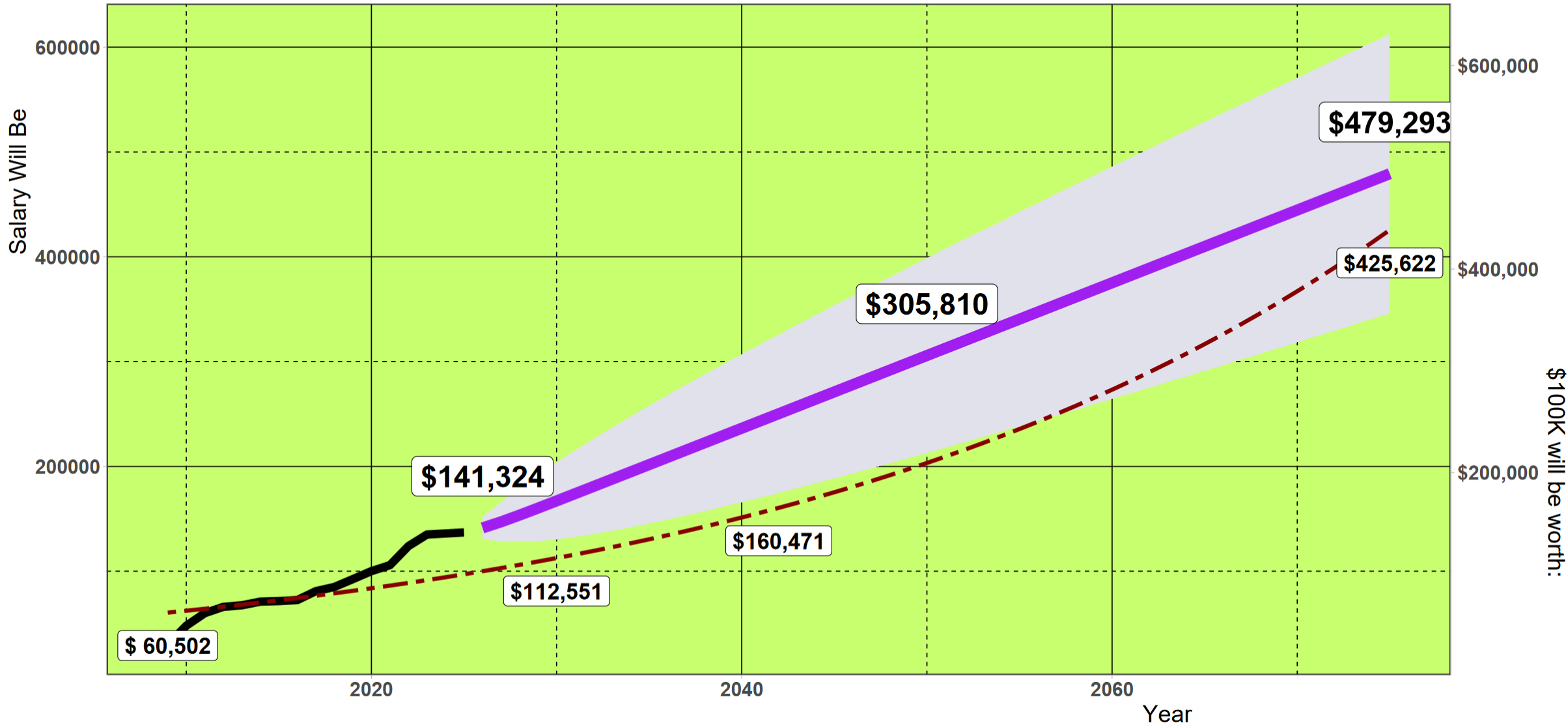


# Salary Forecast Practice



# The following code to create a dataframe and remove duplicated rows is always executed and acts as a preamble for your script:

```
# dataset <- data.frame(Salary, Year)
# dataset <- unique(dataset)
```

# Paste or type your script code here:

```
options(scipen=999)
library(forecast)
library(tseries)
library(ggplot2)
salaryhistory <- dataset
#makes time series:
salaryhistorytimeline=ts(salaryhistory$Salary,start = min(salaryhistory$Year),end=max(salaryhistory$Year),frequency=1)
#this is where it picks the p, d, and q for the arima:
salaryfuturemodel=auto.arima(salaryhistorytimeline,ic="aic",trace=TRUE)
salaryhistorytimeline=ts(salaryhistory$Salary,start = min(salaryhistory$Year),end=max(salaryhistory$Year),frequency=1)
salaryfuturemodel=auto.arima(salaryhistorytimeline,ic="aic",trace=TRUE)
#95% confidence level and h = time horizon. 1 above = year so this is 50 years.
mysalaryforecast=forecast(salaryfuturemodel,level=c(95),h=50)
```

#need to get historic data

```
df_salaryhistory <- data.frame(
  Year = as.numeric(time(mysalaryforecast$x)),
  Salary = as.numeric(mysalaryforecast$x),
  Type = "Historical"
)
```

#do the forecast

```
df_forecastpart <- data.frame(
  Year = as.numeric(time(mysalaryforecast$mean)),
  Salary = as.numeric(mysalaryforecast$mean),
  Lower = as.numeric(mysalaryforecast$lower[, "95%"]),
  Upper = as.numeric(mysalaryforecast$upper[, "95%"]),
  Type = "Forecast"
)
```

#SALARY WILL BE DATA LABELS:

```
print(n)
```

```

#to make dataframe for datalabels. AND I copied this part from AI since I didn't know how to do it :-)
#APPARENTLY IT starts at column one, then counts every row and divides it by two then counts every row
# so you end up with 1, 25, and 50 in this example. Then that gets used for the data tables
df_labels <- df_forecastpart[c(1, round(nrow(df_forecastpart)/2), nrow(df_forecastpart)), ]
# 2. Add rounding to the Salary values so the labels look clean (e.g., $65,320 instead of $65320.123)
df_labels$LabelText <- paste0("$", format(round(df_labels$Salary, 0), big.mark = ","))

#inflation line (apparently its easier to calculate it in R than pull it in from a Power BI table)
Inflation_rate <- 0.03
Base_year = 2026
base_salary <- 100000
start_year <- min(df_salaryhistory$Year)
end_year <- max(df_forecastpart$Year)
all_years <- start_year:end_year

#inflation "data frame"
df_inflation_line <- data.frame(
  Year = all_years,
  InflationSalary = base_salary * (1 + Inflation_rate)^(all_years - Base_year)
)

#Inflation DataLabels
df_labels_two <- df_inflation_line[c(1, round(nrow(df_inflation_line)/3), round(nrow(df_inflation_line)/2), nrow(df_inflation_line)), ]
df_labels_two$LabelTexttwo <- paste0("$", format(round(df_labels_two$InflationSalary, 0), big.mark = ","))

p <- ggplot()+
  geom_ribbon(data = df_forecastpart, aes(x = Year, ymin = Lower, ymax = Upper), fill = "#E3E3EE", linewidth = 0.3)+
  geom_line(data = df_salaryhistory, aes(x = Year, y = Salary), size = 3, color = "black")+
  geom_line(data = df_forecastpart, aes(x = Year, y = Salary), size = 4, color = "purple")+
  geom_line(data = df_inflation_line, aes(x = Year, y = InflationSalary), size = 1.5, color = "darkred", linetype = "twodash")+

```

```

ggtitle(expression(~"*underline('Salary Forecast Practice')"))+
  xlab("Year")+
  ylab("Salary Will Be")+
  theme_light()+
  theme(
    panel.background = element_rect(fill = "darkolivegreen1", color = "black", linewidth = 1),
    plot.background = element_rect(fill = "white"),
    plot.title = element_text(hjust = 0.5, size = 30, face = "bold"),
    axis.title.y = element_text(hjust = 0.8, size = 18),
    axis.title.x = element_text(hjust = 0.8, size = 18),
    axis.text.y = element_text(size=15, face = "bold"),
    axis.text.x = element_text(size=15, face = "bold"),
    panel.grid.major = element_line(color = "black", linewidth = 0.4),
    panel.grid.minor = element_line(color = "black", linewidth = 0.4, linetype = "dashed")
  )+

  scale_y_continuous(

    sec.axis = sec_axis(
      trans = ~. * (1+Inflation_rate),
      name = "$100K will be worth: ",
      labels = scales::dollar
    )
  )+

  geom_label(
    data = df_labels,
    aes(x = Year, y = Salary, label = LabelText),
    fontface = "bold",
    color = "black",
    vjust = -0.8,      # Nudges the label slightly above the line
    size = 8           # Adjusts text size
  )+

  geom_label(
    data = df_labels_two,

```